

Biochemical Guardrails — Q-Twin

Draft prepared from the 45-chip raw dataset (14/15/20/21 Mar 2026). This is a first pass built directly off the computed numbers — Eva should read through, correct anything that conflicts with lab notes, and own the final judgment calls (flagged below).

Errata (fixed since the first draft): Delta-f for the probe/target stages was originally computed as each stage's own start-to-end drift. That was wrong — cross-checking against the lab's own "All results_PCA3.xlsx" (Mean&SE sheet) showed the correct formula is a stage's ENDPOINT frequency minus the PRIOR stage's endpoint frequency (e.g. probe-endpoint minus chi-endpoint). ingest_raw_curves.py has been fixed and this now reproduces the -62.96 Hz / 10 μ M reference almost exactly (see Section 2). This also changed which chips are SUCCESS vs FAILURE.

Update: chip 15Mar_No.16 is now marked EXCLUDED, not FAILURE (~20,600 Hz nonsensical jump between its CHI and probe stages, a lab-flagged bad measurement, absent from the lab's own Success-rate tally too). 44 of 45 chips are valid: 29 SUCCESS, 15 FAILURE (updated again after Eva's Q4 threshold change in Section 3 — was 30/14 under the old 0 Hz cutoff). The kinetic biomarker (Section 5) has also been redefined as early displacement, replacing the original slope version which was confirmed not predictive — see Section 5(c), and a replicate-concordance rule was added for the rate biomarker — see Section 5(d).

1. Concentration bounds

Valid range: 0.5–10 μ M. Concentrations outside this range are not physically meaningful for this assay and should be excluded from training data for the AI models in Weeks 2+.

DRAFT — needs Eva's review: this range is taken directly from the Week 1 guide (Task 5, item 13). The raw dataset also includes 20 μ M, 40 μ M and NC chips, used here only to characterize the crowding inversion (Section 2) and negative-control behavior (Section 4) — confirm this is the intended scope, not a contradiction.

2. The 10 μ M crowding inversion

Above roughly 10 μ M (probe concentration during the probe-deposition stage), Δf stops getting more negative and starts trending positive — surface crowding limits further mass loading, and non-specific/multilayer effects can push the frequency back up instead of down.

Reference number from the published paper/summary: -62.96 Hz at 10 μ M, room temperature.

Concentration	Chips (n)	Mean probe Δf (Hz)	Range (Hz)	Note
0 μ M	3	+10.40	+6.28 to +15.65	baseline, no probe
5 μ M	3	-11.58	-26.05 to +7.01	
10 μ M	3	-62.04	-105.86 to -31.61	peak binding
20 μ M	3	-55.30	-120.81 to -10.36	above crowding onset
40 μ M	3	-43.60	-76.77 to -10.07	

Table computed from the 14 Mar concentration-sweep chips (probe-endpoint minus chi-endpoint, averaged per chip; n=3 per concentration). The 10 μ M mean, -62.04 Hz, now matches the -62.96 Hz reference almost exactly (the ~1 Hz residual is well within instrument noise / endpoint-selection rounding). The concentration trend is also physically clean: strongly positive at 0 μ M (no probe, no mass added), peak binding at 10 μ M, then decreasing magnitude at 20-40 μ M — consistent with the crowding-inversion story, though note it never goes positive again at 20-40 μ M in this dataset (magnitude just shrinks), so 'inversion' may be too strong a word for what's really 'crowding-limited plateau'.

This resolves clarifying_questions.md item 2 — the mismatch was a bug in the original Δf computation (see Errata at top), not a data or lab-notes issue. Fixed in ingest_raw_curves.py.

DRAFT — needs Eva's review: Eva's Q2/Q3 messages also state the Target Hybridization stage has a positive +63.49 Hz shift at 10 μ M (vs the probe stage's -62.96 Hz), with crowding described as a gradual onset. Checked against this dataset's Δf_{target} at 10 μ M (n=4, noisy): +85.325, -211.450, +42.855, +0.120 Hz — mean -20.79 Hz, or +42.77 Hz excluding the -211.45 outlier. Neither cleanly matches +63.49 Hz. Not written into this table as confirmed, since (unlike the probe-stage number) there isn't enough data here to independently pin it down — see clarifying_questions.md item 15. Eva: could you point to the specific sheet/computation +63.49 Hz comes from?

3. Chip failure logic — threshold updated per Eva's Q4

SUCCESS if probe-stage $\Delta f \leq -0.5$ Hz. FAILURE otherwise ($\Delta f > -0.5$ Hz, or missing probe data). Δf here is probe-endpoint minus chi-endpoint (see Errata). The -0.5 to 0 Hz band is treated as within instrument baseline noise, not a real signal.

Across the 44 valid chips (15Mar_No.16 excluded): 29 SUCCESS, 15 FAILURE (65.9% success rate). This moved one chip relative to the old 0 Hz cutoff: 21Mar_No.9 (-0.18 Hz) flips from SUCCESS to FAILURE, since it falls inside the new noise buffer. Mean probe Δf for SUCCESS chips is -78.56 Hz; mean for FAILURE chips is +19.93 Hz.

This resolves clarifying_questions.md item 4/11 — Eva confirmed -0.5 Hz as the buffer. Note the 65.9% chip-level rate is a bit further from the teacher's 70.18% (40/57) than the old 0 Hz cutoff's 68.2% was — see item 1, still most likely a chip-count vs trial-count denominator difference, not a real disagreement.

4. Real negative-control behavior

Chips No.7 and No.29 (21 Mar) are the only genuine 'no target present' runs. Plots: figures/NC_21Mar_No_7.png and figures/NC_21Mar_No_29.png.

With the corrected Δf , both NC chips are clearly FAILURE and clearly positive: No.7 is +3.29 Hz, No.29 is +22.615 Hz (previously this looked like a near-zero borderline call at +0.08 Hz under the buggy formula — the fix makes the negative-control result much cleaner and more convincing, not less). Draft description of the curves themselves: both NC chips show flat, low-amplitude, monotonic upward drift over their full run (roughly +10-15 Hz over 150-220s) with no sharp steps or large excursions — this looks like slow thermal/baseline drift, not a binding event. By contrast, a genuine hybridization curve (e.g. chip No.2, a SUCCESS chip from the same 21 Mar session) shows a fast initial drop of tens of Hz within the first ~20-30s and sharp step-like jumps later in the run, both absent from the NC curves. Chip No.29's probe-stage replicates are noisier than No.7's, with small dips instead of a smooth drift, but neither shows the large negative excursion characteristic of a SUCCESS chip.

Update: Eva has confirmed this is lab-verified thermal/buffer baseline drift, not binding (Q5) — the paragraph above is no longer just Evin's inference from the plots.

5. The Novel Kinetic Biomarker — REDEFINED as displacement, validated

(a) Time window

30 seconds, measured from the start of the probe-stage curve (also checked at 15s/45s/60s, see (c)).

(b) Definition — changed from slope to displacement

The original definition was the SLOPE of a linear fit to the probe curve's first 30s (binding_rate_probe_dfdt_30s in chip_summary.csv). That version is confirmed NOT predictive of outcome (see prior draft in version history / clarifying_questions.md item 8) — it's a same-stage-only quantity, mechanically disconnected from Δf_{probe} , which is a cross-stage quantity (see Errata).

Redefined as early DISPLACEMENT: the probe curve's own value AT $t=30s$ (linearly interpolated) minus the CHI-stage endpoint baseline — the same cross-stage baseline Δf probe itself uses, just read 30s into the run instead of at the end. New column: `early_displacement_30s` in `chip_summary.csv`.

(c) Validation

Correlation between `early_displacement_30s` and the corrected Δf probe is 0.9997 across the 44 valid chips — this is essentially an early readout of the same signal, which is exactly why it works as a feature (unlike the slope version, which was a genuinely different quantity). Simple threshold classification (predict SUCCESS if $\text{displacement} < 0$) scores 97.5% (39/40 scoreable chips) at the 30s window in this implementation, and holds flat at 97.5% across 15s/30s/45s/60s windows too.

These numbers were independently re-derived, not copied from Eva's message — the exact figures (97.5% here vs Eva's reported 97.7%/15s, 95.5%/30s, 93-98% range) differ slightly, most likely from a difference in interpolation or accuracy-scoring method between the two implementations. Both independently land on the same conclusion: displacement is a strong, usable feature; slope was not. Worth a quick sync between the two methodologies before Week 2, but not blocking.

The one thing to watch: chip 21Mar_No.29 (true negative control) still shows an early value consistent with weak binding under the old slope metric, but the corrected full-run Δf (+22.615 Hz) is unambiguous FAILURE, and the new displacement metric agrees with the outcome for this chip. The previously-flagged single-replicate noise issue was specific to the slope calculation, not the displacement one.

(d) Replicate concordance rule for the rate biomarker (Q7)

Per Eva's spec: the rate-based biomarker (`binding_rate_probe_dfdt_30s`) is only averaged across replicates if they agree (same sign AND relative difference $|a-b|/\max(|a|,|b|) \leq 0.5$). If they diverge, the chip is flagged `binding_rate_replicate_status = DIVERGENT_REPLICATES` and the rate is left null rather than passing a corrupted average downstream. New column in `chip_summary.csv`.

Verified against Eva's own example: chip 21Mar_No.29's replicate rates are -0.112 Hz/s (R1) vs -0.029 Hz/s (R2) -- same sign but a ~74% relative gap -- correctly flagged `DIVERGENT_REPLICATES`. Worth noting: the displacement metric in (b)/(c) does NOT diverge for this same chip (19.08 vs 22.33 Hz) because R2's spike happens after the 30s window -- the divergence is specific to the early-rate calculation.

DRAFT — needs Eva's review: with this 0.5 tolerance, 14 of the 22 chips that have 2+ probe replicates (64%) get flagged divergent, not just No.29. That's either genuinely high replicate-to-replicate noise across this instrument, or the tolerance is too strict for real QCM data. Eva/the teacher should spot-check a handful of the other 13 flagged chips before this tolerance is hard-coded into Week 2's generator/gatekeeper logic. See `clarifying_questions.md` item 12.

Sources

- Computed values: `qtwin/data/chip_summary.csv` (45 chips, all 4 sessions).
- Cross-check against Eva's `chip_index.csv`: `qtwin/data/cross_check_report.csv` (0 mismatches).
- Plots: `qtwin/figures/` (NC_21Mar_No_7.png, NC_21Mar_No_29.png, and grouped condition plots).